Lab-3: Requirements Engineering & Elicitation

1. Title of the Project

Municipal Complaint Management System (MCMS)

2. Overview of Requirements Engineering

Requirements Engineering is the process of identifying, analyzing, documenting, and validating the services a software system must provide along with the constraints under which it will operate.

The primary goals are:

- Understanding real civic problems faced by citizens and municipal staff.
- Discovering explicit and hidden requirements.
- Ensuring consistency, completeness, and feasibility before design and implementation.

3. Requirement Elicitation Process & Techniques Used

To gather comprehensive requirements, multiple elicitation techniques were conducted:

3.1 Stakeholder Interviews

- **Target Audience:** Ward Supervisors, Municipal Desk Officers, Field Workers, and Citizens.
- **Findings:**
    - Officers highlighted repeated complaints and lack of clear photographic proof.
    - Field workers stated they need clear location details and category classification.
    - Citizens demanded a simple tracking mechanism without visiting the ward office repeatedly.

3.2 Surveys & Questionnaires

- Distributed an online and on-ground questionnaire to citizens across multiple wards.
- **Key Insights:**
    - 85% of citizens preferred logging complaints online instead of visiting an office.
    - 72% demanded a photo upload option for evidence.

    - 90% stated tracking complaint progress in real time is the most critical feature.

3.3 Document & Existing System Analysis

- Studied the existing manual paperwork registers and municipal standard operating procedures (SOPs).
- Identified gaps: lack of SLA enforcement, untracked grievance delays, and absence of an escalation matrix.

4. Stakeholder Identification

- **Citizens:** Individuals living within municipal limits who report civic issues (sanitation, broken roads, faulty water pipelines, defunct streetlights).
- **Municipal Officers (Supervisors/Admins):** Officials responsible for monitoring, validating, and delegating complaints to specific ground personnel.
- **Field Workers (Ground Staff):** Sanitation teams, electricians, and road repair workers who physically resolve problems on-site.
- **System Administrator:** Manages user roles, ward master data, department configurations, and system security.

5. Elicited Requirements

5.1 Functional Requirements (FR)

- **FR-01 (User Registration & Authentication):** Citizens and staff shall be able to register and log in securely using credentials.
- **FR-02 (Complaint Lodging):** Citizens shall be able to submit a complaint by selecting a department/category, typing a description, adding location/address, and uploading photo evidence.
- **FR-03 (Unique Tracking ID Generation):** The system shall auto-generate a unique Tracking ID upon successful complaint submission and notify the citizen.
- **FR-04 (Automated Routing & Assignment):** The system shall route complaints to the appropriate department (e.g., Sanitation, Roads, Electricity, Water Board) and allow officers to assign them to designated field workers.
- **FR-05 (Status Tracking):** Citizens shall be able to track live ticket progress using their Tracking ID (`Submitted` -> `Assigned` -> `In-Progress` -> `Resolved`).
- **FR-06 (Field Resolution & Proof):** Field workers shall view allocated tasks and mark them resolved after completing the physical task.
- **FR-07 (Citizen Feedback & Rating):** Citizens shall be able to provide feedback and a rating once the grievance is marked as resolved.
- **FR-08 (Escalation Matrix):** Unresolved grievances crossing a defined SLA timeframe (e.g., 7 days) shall automatically escalate to higher administrative authorities.

5.2 Non-Functional Requirements (NFR)

- **Performance:** The portal shall respond to user requests within 2 seconds under standard network conditions.
- **Security:** User credentials and passwords must be securely hashed. Only authorized users with appropriate roles can access admin and worker dashboards.
- **Availability & Reliability:** The system shall maintain 99% uptime so complaints can be registered 24/7.
- **Usability:** The citizen interface shall be lightweight, mobile-responsive, and require minimal technical literacy.
- **Scalability:** The database and application architecture shall support concurrent submissions during emergency civic scenarios (e.g., heavy rains or water contamination).